LIBRARIES
reads or Sanger traces

AID-competent animals

AID-null controls (two or more)

nanopore · Illumina · PacBio · Sanger

Sample sheet

file	animal	control
a.fastq	Wt 2	no
b.fastq	KO 1	yes
c12.ab1	Wt 3	no

labels stay with every flag

READ PREPARATION
every read is gated

Gates

too long, or > 1% errors/base
Sanger: Q ≥ 20 core kept

Competitive alignment

target ✓ · paralogue · other

each read goes to its best match; a paralogue can be a target of its own

Read accounting

used · dropped

each drop with its reason, every library (Figure S2)

AUTOMATED ANALYSIS
calibrated by the AID-null controls

Background

per position, change, strand and quality,
+ a leave-one-out margin

Candidate test

observed

Poisson tail against the background; in ≥ 50% of reads: germline, set aside

Threshold set by the controls (target–decoy)

kept · p-value →

calls (filled) against decoys: each control called against the others (open)

threshold: predicted false calls ≤ 5% of calls kept

Sanger: Q ≥ 20 changes counted; background shown beside

Bookkeeping checks (reported, never applied)

copied trace · clone in two animals · mutated control · Wt 2 ≠ Wt 4 label mismatch · load outlier

RESULTS
per animal, then per group

Mutation profiles

density · spectrum · hotspots · position · clones, trees · indels

Group tests

Welch t on $\log_{10}$ density, one value per animal

case · control

● mutation ● sequencing error